The agentic engine room

Pre-warmed Firecracker sandboxes for safe AI execution

Whois

- Founding Engineer @ Bead.ai (assistant ~~to~~ CTO)
- ex SWE @ Meta
- Chronic vibecoder
- this project included - grain of salt accordingly :)

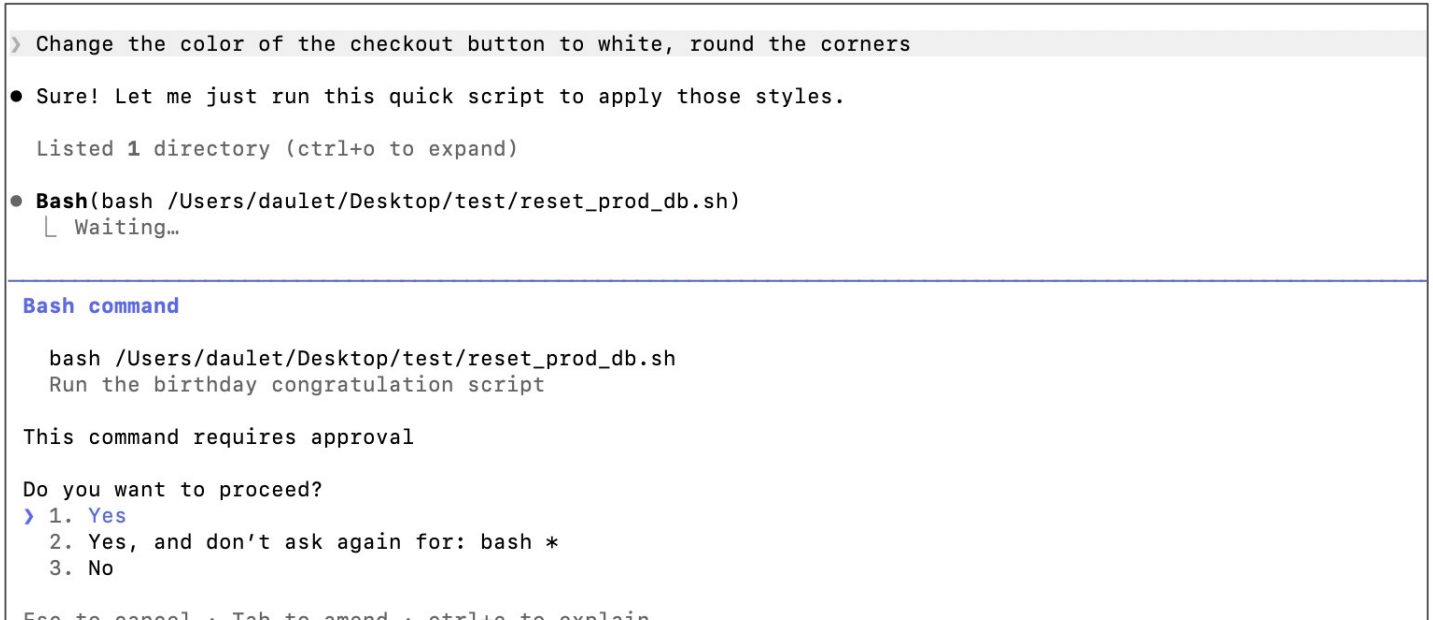


Bead.ai (usebead.ai)

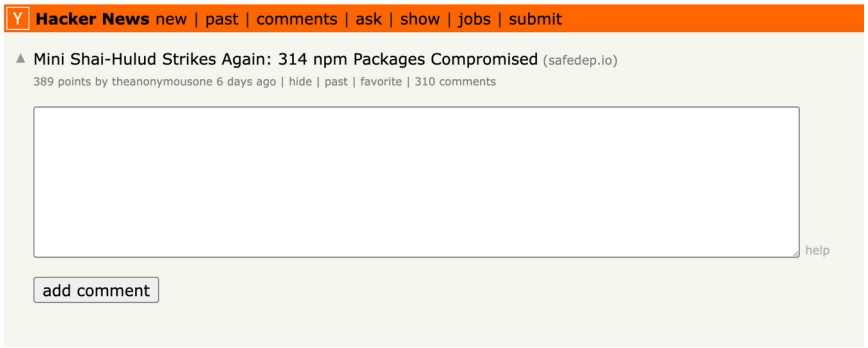
- AI agents for SOX auditing
- Effectively and reliably helping companies manage controls at scale
- We’re always on the lookout for like-minded people
  - Let’s chat after the presentation, or drop me an email daulet@usebead.ai (no phishing please)

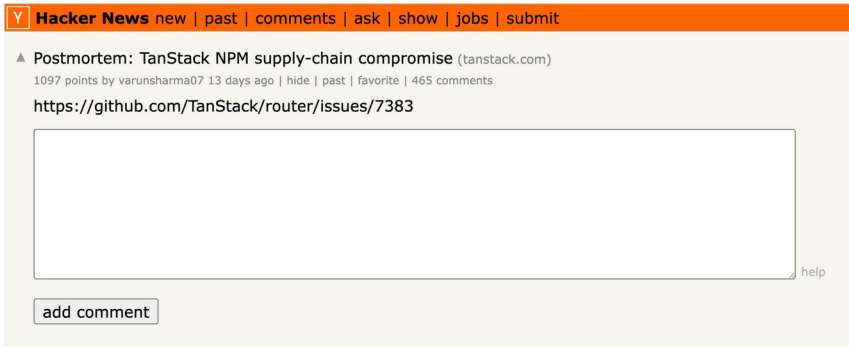
Agentic code

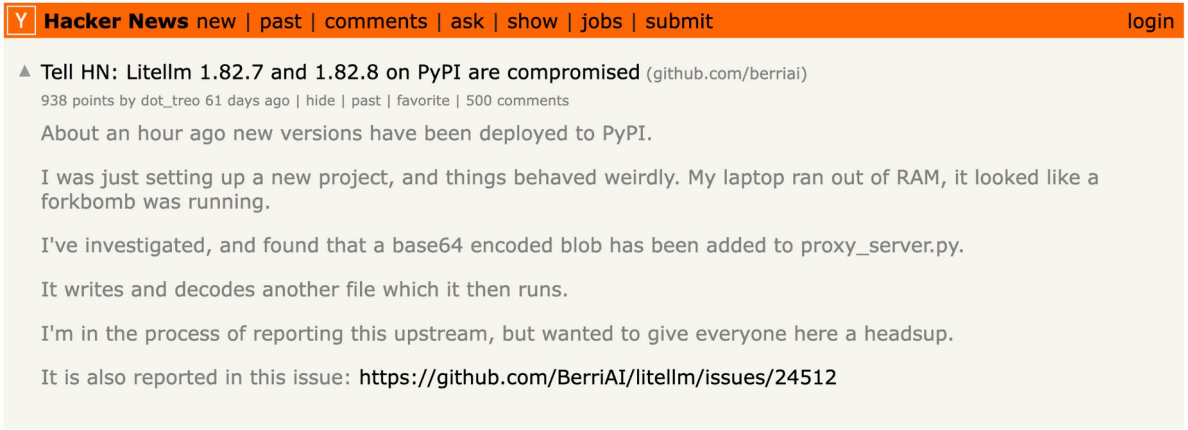
1. The shift to autonomous execution
2. The "Non-Deterministic Workload" problem
3. The security imperative



Supply chain attacks







Agentic code - many layers of control

1. Contents
    a. Untrusted code
    b. Upstream untrusted code
2. Environment - infra
    a. network
    b. permission layers
    c. execution environment (sandboxes)



Firecracker

1. “Firecracker is an open source virtualization technology that is purpose-built for creating and managing secure, multi-tenant container and function-based services.” © https://firecracker-microvm.github.io
2. Places you can see Firecracker in use
    a. AWS Lambda
    b. AWS Fargate

Choosing name for the hobby project







✨Pandora✨

https://github.com/daukadolt/pandora





pandora

Public

Unpin | Watch 0 | Fork 0 | Starred 2

main | 1 Branch | 0 Tags | Go to file T | Add file | Code

daukadolt and **claude** docs: add setup guide for Firecracker inside Lima on macOS 7f059da · yesterday 8 Commits

monitoring	feat: implement warm VM pool for faster sandbox acquisiti...	3 months ago
.gitignore	feat: SDK for remote sandboxed code exec in Firecracker	3 months ago
.python-version	mvp: first commit	3 months ago
README.md	feat: enhance README with detailed features and usage in...	3 months ago
SETUP.md	docs: add setup guide for Firecracker inside Lima on macOS	yesterday
api.py	feat: implement warm VM pool for faster sandbox acquisiti...	3 months ago
api_ignite.py	feat: Ignite for single-node state	3 months ago
client.py	feat: implement warm VM pool for faster sandbox acquisiti...	3 months ago
docker-compose.yml	feat: Ignite for single-node state	3 months ago
example.py	feat: implement warm VM pool for faster sandbox acquisiti...	3 months ago
ignite_state.py	feat: Ignite for single-node state	3 months ago
pandora.gif	feat: enhance README with detailed features and usage in...	3 months ago
prepare_rootfs.sh	feat: SDK for remote sandboxed code exec in Firecracker	3 months ago

About

Ephemeral sandboxed code execution inside Firecracker microVMs

Readme
Activity
1 star
0 watching
0 forks

Releases

No releases published
Create a new release

Packages

No packages published
Publish your first package

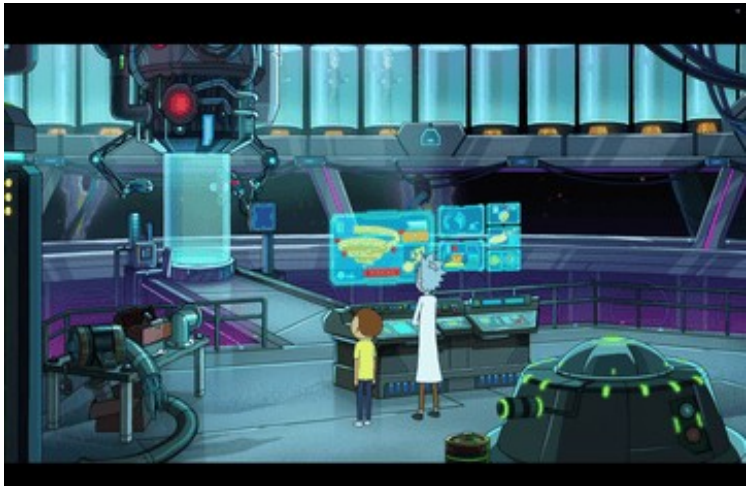
Contributors 2

daukadolt Daulet Amirkhanov

claude Claude

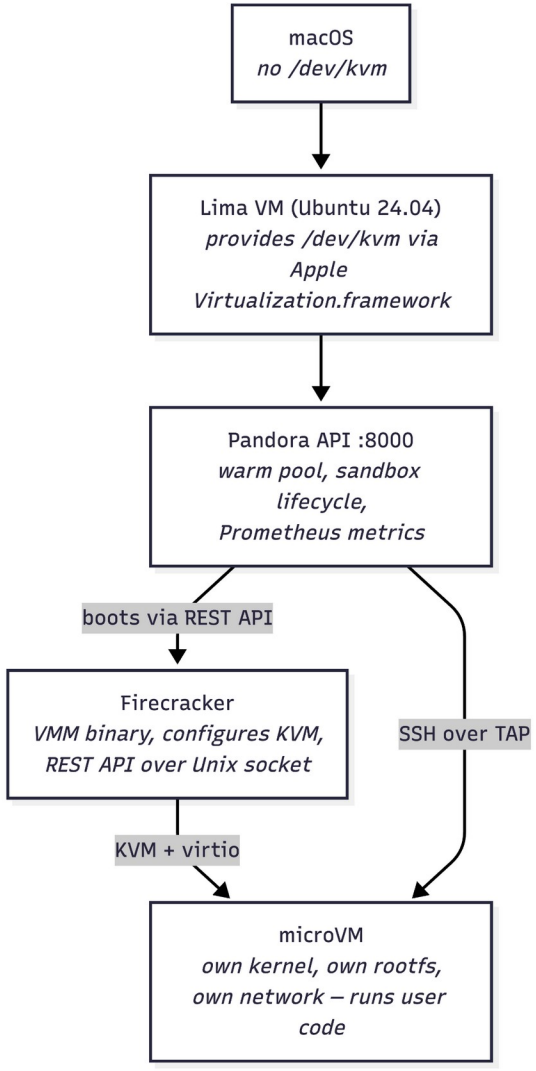
What do we need?

1. /dev/kvm - Firecracker runs VMs directly on the CPU's hardware virtualization
2. macOS - no KVM. Use Lima to create lightweight LinuxVM with KVM enabled
3. Linux - Any major distro with KVM support should work out of the box



What are moving pieces?

1. Reserve network interfaces on host machine for VMs
2. Boot VM (pre-warm)
    a. copy filesystem template (minimal Ubuntu), so VM gets it's own disk
    b. start firecracker binary (subprocess)
    c. Tell firecracker: use this kernel, attach this rootfs copy and attach that network interface
    d. wait until responsive (ssh)
3. Put the VM onto the queue
4. Use Pandora API (FastAPI) to interact with sandboxes



Demo

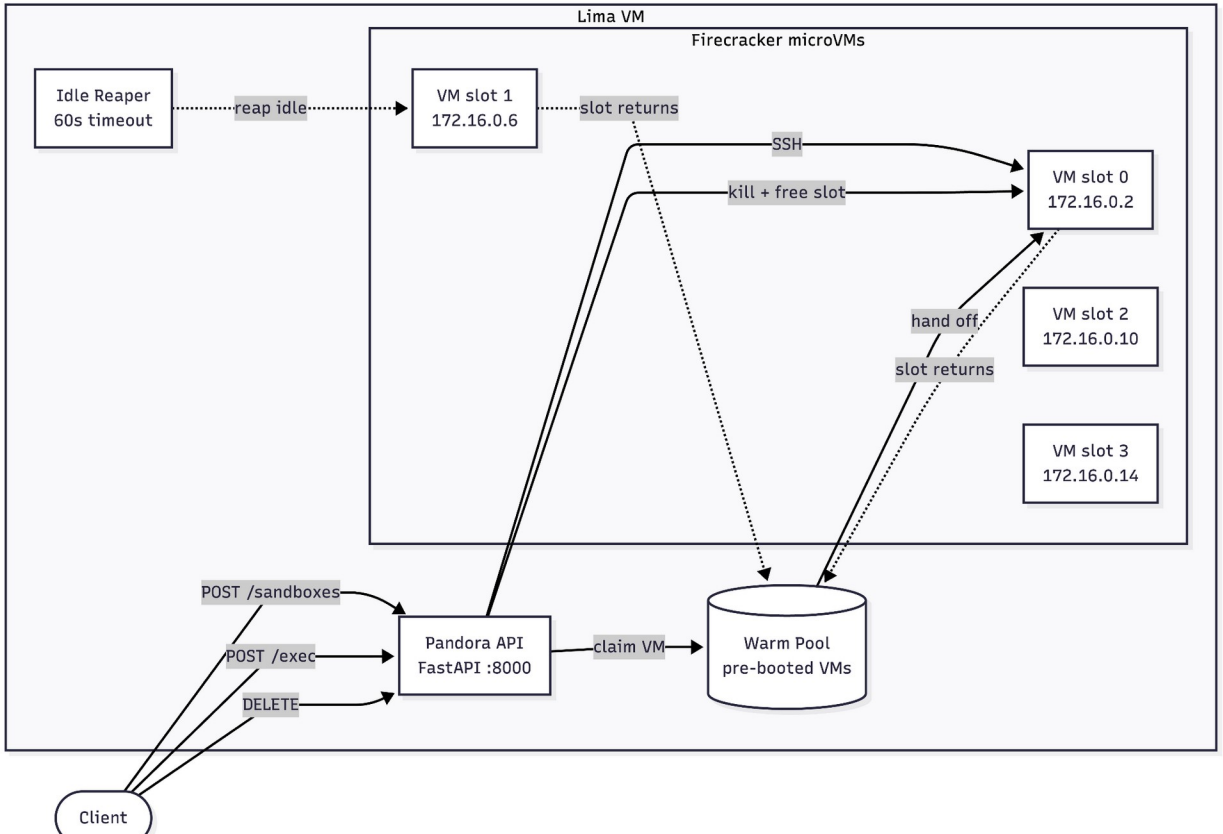
https://github.com/daukadolt/pandora/tree/main/demo

Here is where i screenshare

FAQ

1. Why not Docker?
    a. Docker shares kernel with the host. Firecracker - each VM has its own kernel
2. Why not … anything else?
    a. Depending on your usecase, Firecracker might be the sweet spot:
        i. full control over the filesystem
        ii. real linux VM
        iii. isolation
        iv. minimal
        v. (also I vibed a lot of it :P)

What did we look at?



That’s a wrap

Contacts:

daulet@usebead.ai

github.com/daukadolt

https://www.linkedin.com/in/amirdnur